TECH BIZ SECURITY · SOC ANALYST INTERNSHIP · BATCH B-503

WEEK 4 · TASK 10

What Is a SIEM? & Setting Up Splunk

Submitted by	**Oscar ALIDJINOU**
Student ID	**TBS-INT-2026-2398**
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01 Objective

In my first weeks I read logs one file at a time: auth.log on Linux, Event Viewer on Windows, Apache access logs, Wireshark captures. A real company produces millions of such events every day from hundreds of systems, and no analyst can read them by hand. A SIEM is the tool that solves this. This report explains what a SIEM does and why it matters, describes how I installed Splunk in my lab, and shows how it fits into a SOC analyst's daily work.

02 Proof of Completion

Room / activity	Duration	Tasks	Status
TryHackMe: Introduction to SIEM	120 min	7	Completed, room progress 100%
Lab: installing Splunk	-	-	Installed and running

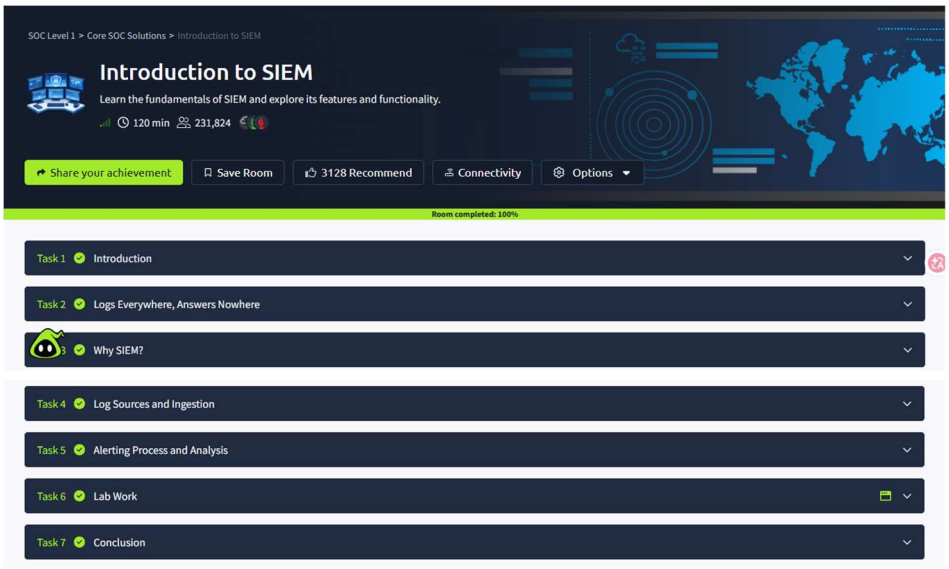


Figure 1. Introduction to SIEM: all 7 tasks completed (100%).

The room's practical lab

The last task put me in front of a simulated SIEM. I read its event table, made up of Windows process-creation events (Event ID 4688), found the event that had caused an alert, and worked out which user, which host and which rule term were involved. This is the daily routine of a Tier 1 analyst.

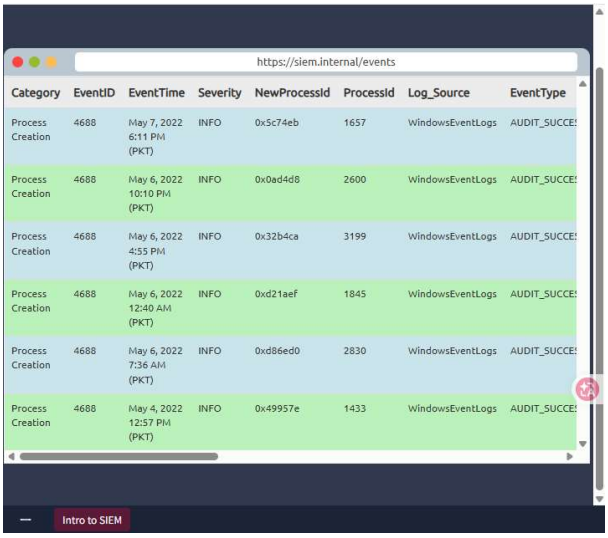


Figure 2. The simulated SIEM event table from the room's lab.

03 What Is a SIEM?

SIEM stands for Security Information and Event Management. It is a security platform that collects logs from many different sources, converts them into one common format, correlates them, and raises alerts when they match a detection rule. It is the core tool of a SOC analyst.

3.1 The problem it solves: logs everywhere, answers nowhere

Every device in a network records what happens on it. These devices are called log sources. Without a SIEM, the logs exist but nobody can use them effectively:

Problem	What it means in practice
Too many log sources	Servers, workstations, firewalls, routers and applications each produce their own logs, far too many to read by hand.
No centralisation	Logs sit on each device, so an analyst must log in to machine after machine to follow an incident.
Limited context	One log line on its own rarely tells the full story; it needs other sources to be understood.
Limited analysis	Searching raw files manually is slow and misses patterns spread over several systems.
Different formats	Each system writes in its own format, so the same fact, such as a username, is hard to compare.

3.2 What a SIEM does

Function	What it does
Centralised log collection	Gathers logs from every source into a single platform.
Normalisation	Converts different formats into one consistent structure, so that a field such as source IP means the same thing everywhere.
Correlation	Links events from different sources to reveal an attack that no single log shows.
Real-time alerting	Raises an alert when events match a detection rule.
Dashboards and reporting	Shows a summary of the environment and produces reports for management and audits.

3.3 Three benefits of a SIEM

#	Benefit	Why it matters
1	One place to see everything	Centralised, normalised logs let an analyst search the whole company at once instead of machine by machine, which saves hours during an investigation.
2	Faster detection	Correlation and real-time alerts spot attacks as they happen. For example, many failed logins followed by a success is something I found by hand in Week 2; a SIEM detects it automatically.
3	Investigation and compliance	Stored, searchable history supports incident response and forensics, and ready-made reports help to prove that rules and regulations are followed.

> **A link to my earlier work**
> Almost everything a SIEM automates is something I have already done manually: reading failed logins (Event ID 4625 and auth.log), spotting a flood of 404 responses, and comparing against a baseline. A SIEM does it continuously, on every system at once.

04 Log Sources and How Logs Reach the SIEM

4.1 Log sources

Source type	Examples	What it records
Windows host	Event Viewer: Security, System, Application logs.	Logons (4624, 4625), process creation (4688), account changes (4720).
Linux host	/var/log/auth.log (or secure), cron, kern, syslog.	Logins, sudo, scheduled jobs, kernel events.
Web server	/var/log/apache2 or /var/log/httpd (access and error logs).	Every request and response, useful for spotting web attacks.
Network devices	Firewalls, routers, IDS/IPS, DNS servers, VPN.	Connections allowed or blocked, alerts, DNS lookups.

4.2 Ingestion methods

Method	How it works
Agent / forwarder	A small program installed on the machine reads the important logs and sends them to the SIEM server. Splunk calls it the Universal Forwarder; Wazuh calls it the agent.
Syslog	A widely used protocol that sends log messages from systems such as servers and network devices to a central destination in real time.
Manual upload	A file of offline logs is uploaded for quick analysis, for example during an investigation.
Listening port	The SIEM listens on a chosen port and devices send data to it.

05 Installing Splunk in My Lab

I chose Splunk because it is the SIEM most often met in job descriptions and because its search language is useful on its own. Wazuh is the main open-source alternative. I installed Splunk Enterprise (version 10.6.1.1) on an Ubuntu virtual machine.

	Splunk	Wazuh
Type	Commercial product with a free trial and a limited free licence.	Fully open source.
Strength	Very powerful search and reporting with its SPL query language.	Built-in endpoint monitoring and agents, with no licence cost.
Why I chose it	Widely used in industry and well suited to learning search.	A good second option to try later.

5.1 Steps

#	Step	What I did
1	Download the package	On the Splunk website I chose Linux and the .deb package (about 1.35 GB) and copied its wget link.
2	Download it in the VM	I ran the wget command in the terminal. The file (1.32 GB) downloaded in about 22 minutes.
3	Install the package	I installed the .deb file, which places Splunk in /opt/splunk (the folder visible in the running processes).
4	Start Splunk and enable it at boot	Splunk runs as a systemd service, started automatically at boot, with the management port 8089.
5	Verify the service	systemctl status Splunkd shows active (running).
6	Log in to Splunk Web	I opened the web interface in a browser and signed in as the administrator account created during installation.

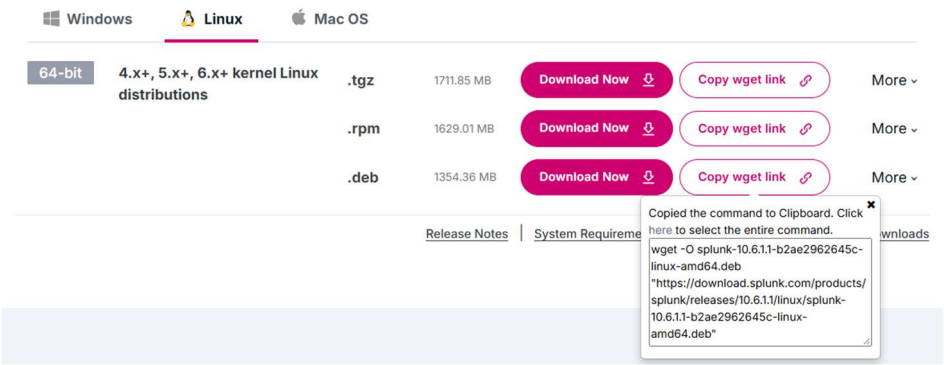


Figure 3. Choosing the Linux .deb package and copying the wget download command.

```
server_soc@ubuntu:~$ wget -O splunk-10.6.1.1-b2ae2962645c-linux-amd64.deb "https://download.s
plunk.com/products/splunk/releases/10.6.1.1/linux/splunk-10.6.1.1-b2ae2962645c-linux-amd64.de
b"
--2026-11-06 01:14:21--  https://download.splunk.com/products/splunk/releases/10.6.1.1/linux/
splunk-10.6.1.1-b2ae2962645c-linux-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 52.84.45.44, 52.84.45.100, 52.84.45.15
, ...
Connecting to download.splunk.com (download.splunk.com)|52.84.45.44|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1420151880 (1,3G) [binary/octet-stream]
Saving to: 'splunk-10.6.1.1-b2ae2962645c-linux-amd64.deb'

splunk-10.6.1.1-b2ae296 100%[===================>]   1,32G  1,64MB/s    in 21m 57s

2026-11-06 01:36:20 (1,03 MB/s) - 'splunk-10.6.1.1-b2ae2962645c-linux-amd64.deb' saved [14201
51880/1420151880]
```

Figure 4. Downloading the Splunk package with wget.

5.2 Checking that it works

The service status confirms Splunk is running. The main splunkd process is up, it is listening on management port 8089, and the checks Splunk runs at startup (critical directories, indexes and the file system) all finished as Done.

```
server_soc@ubuntu:~$ sudo systemctl status Splunkd
● Splunkd.service - Systemd service file for Splunk, generated by 'splunk enable boot-start'
     Loaded: loaded (/etc/systemd/system/Splunkd.service; enabled; vendor preset: enabled)
     Active: active (running) since Fri 2026-11-06 02:15:59 UTC; 15s ago
    Process: 64673 ExecStartPost=/bin/bash -c chown -R splunk:splunk /sys/fs/cgroup/system.s>
   Main PID: 64672 (splunkd)
      Tasks: 9 (limit: 7527)
     Memory: 60.0M (max: 6.2G available: 6.1G)
        CPU: 14.703s
     CGroup: /system.slice/Splunkd.service
             ├─64672 splunkd --under-systemd --systemd-delegate=yes -p 8089 _internal_launch>
             ├─64751 "[splunkd pid=64672] splunkd --under-systemd --systemd-delegate=yes -p >
             └─64758 /opt/splunk/bin/python3.9 /opt/splunk/etc/apps/splunk_pipeline_builders>

nov. 06 02:16:07 ubuntu splunk[64728]:         Checking critical directories...        Done
nov. 06 02:16:07 ubuntu splunk[64728]:         Checking indexes...
nov. 06 02:16:07 ubuntu splunk[64728]:                 Validated: _audit _configtracker _dm_>
nov. 06 02:16:07 ubuntu splunk[64728]:         Done
nov. 06 02:16:09 ubuntu splunk[64672]:         Checking filesystem compatibility...  Done
```

Figure 5. systemctl status Splunkd: the service is active (running).

I then opened Splunk Web and logged in. The home page greets the administrator and lists the installed apps, including Search & Reporting (where searches and dashboards are built), Audit Trail and Data Management.

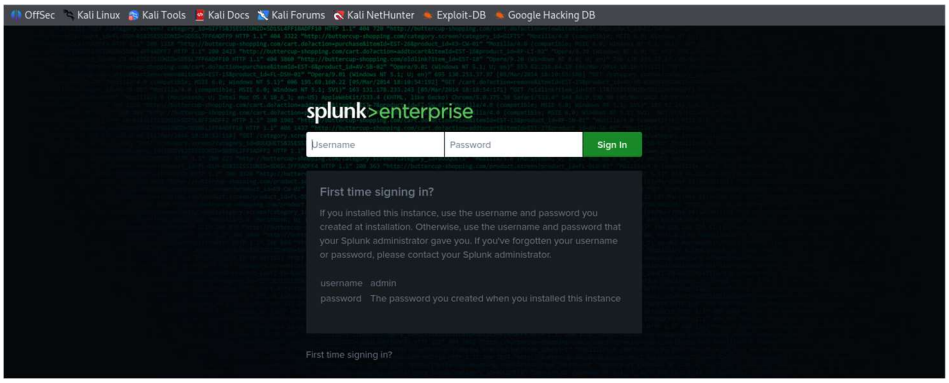


Figure 6. The Splunk Enterprise login page.

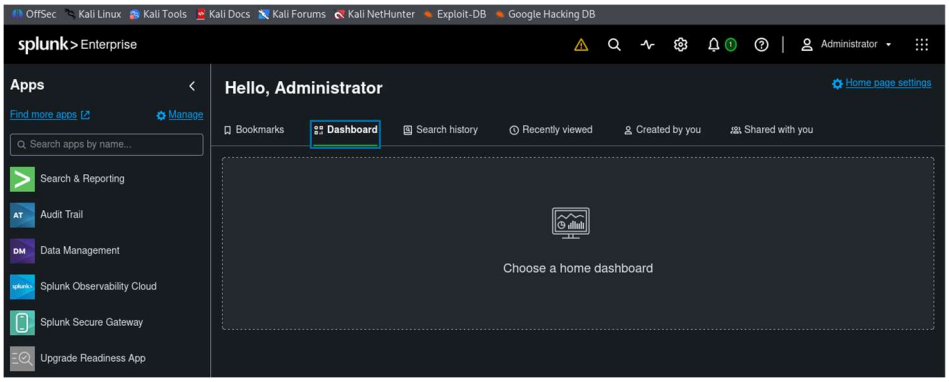


Figure 7. Logged in: the Splunk home page.

06 How a SIEM Is Used in a SOC

6.1 Detection rules

A detection rule is a logical condition that raises an alert when it is true. These examples from the room are typical:

Rule	Alert it raises
5 failed logins within 10 seconds	Multiple failed login attempts (brute force).
A successful login after several failures	Successful login after failed attempts (possible compromise).
A USB device is plugged in, where policy forbids it	Policy violation.
Outbound traffic above a set size, for example 25 MB	Possible data exfiltration.

6.2 Investigating an alert

#	Step	What the analyst does
1	Review the alert	Read which rule fired and which conditions were met.
2	Examine the events	Open the events behind the alert: user, host, process, time, source IP.
3	Decide	Is it a true positive (real threat) or a false positive (harmless)?
4	Act	False positive: tune the rule so it does not repeat. True positive: investigate further, contact the asset owner, isolate the infected host and block the suspicious IP.

> **From my own work**
> The 4625 event I analysed in Week 2 was a mistyped password: a false positive. In a SIEM, I would record that and tune the rule, for example by ignoring one or two failures followed by a success, so the alert does not waste anyone's time again.

07 Reflection and Key Takeaways

- A SIEM does not replace the analyst. It collects, normalises and correlates, so the analyst can spend time on judgement.
- Detection rules are only as good as their tuning: too sensitive and they flood the team with false positives; too loose and they miss attacks.
- Everything I practised by hand in earlier weeks (logs, filters, baselines, event IDs) is what I will now do at scale in Splunk.
- Installing a SIEM myself showed me how it is built: a server, an interface and agents that feed it.

Next step: add my lab's logs to Splunk, build the dashboard above, and write my first detection rule.

08 References

- TryHackMe: Introduction to SIEM; Splunk documentation (installing Splunk Enterprise on Linux); Wazuh documentation.